In the first two problems, we will finally prove the Division Algorithm for non-negative integers.

Problem 1. For $a \in \mathbb{N} \cup \{0\}$, let $P(a)$ be the statement "for all $b \in \mathbb{N}$, there exist $q, r \in \mathbb{N} \cup \{0\}$ such that $a = bq + r$ and $0 \leqslant r < b$."

Prove by strong induction that $P(a)$ is true for all $a \geqslant 0$.

Note. The statement $P(a)$ is a universally quantified statement for b.

Hint. In the inductive step, consider the cases $k + 1 < b$ and $k + 1 \geqslant b$.

Problem 2. Show that the q and r in Problem 1 are unique. That is, show if there also exist q', r' such that $a = bq' + r'$ and $0 \leqslant r' < b$, then $q' = q$ and $r' = r$. You may follow this guide:

1. Prove that $r - r' = b(q' - q)$.
2. Show that $-b < r - r' < b$.
3. Use that if x is an integer multiple of b and $-b < x < b$, then $x = 0$.

Problem 3. Are the following propositions true or false? Justify your conclusion with proof or counterexample. Throughout, suppose that A, B, C, and D are subsets of a universal set U.

1. $A \subseteq B$ if and only if $B^c \subseteq A^c$.
2. The sets $A \cap B$ and $A \backslash B$ are disjoint.
3. If $A \subseteq B$ and $C \subseteq D$ and A and C are disjoint, then B and D are disjoint.
4. If $A \subseteq B$ and $C \subseteq D$ and B and D are disjoint, then A and C are disjoint.
5. If $A \cap C \subseteq B \cap C$, then $A \subseteq B$.
6. If $A \cup C \subseteq B \cup C$, then $A \subseteq B$.

Problem 4. Let A, B, and C be sets. Prove $A \times (B \backslash C) = (A \times B) \backslash (A \times C)$

Problem 5. Let A and B be nonempty sets. Prove that $A \times B = B \times A$ if and only if $A = B$.

Problem 6. For each of the following family of sets $\mathcal{A}$, calculate $\bigcup_{A \in \mathcal{A}} A$ and $\bigcap_{A \in \mathcal{A}} A$. Assume that the universe is $\mathbb{R}$. *You must show that your answer is correct by showing that the given set and your set are equal.*

1. $\mathcal{A} = \{A_z \mid z \in \mathbb{Z}\}$ where $A_z = \left(\frac{2z-1}{2}, \frac{2z+3}{2}\right)$ (Note that this denotes the open interval, not the ordered pair, because the universe is $\mathbb{R}$)
2. $\mathcal{A} = \{A_n \mid n \in \mathbb{Z}_{>0}\}$ where $A_n = \left[0, \frac{1}{n}\right]$.

Problem 7. Let Λ be a nonempty index set and let $\mathcal{A} = \{A_\alpha \mid \alpha \in \Lambda\}$ be an indexed family of sets.

1. Let B be a set such that $B \subseteq A_\alpha$ for all $\alpha \in \Lambda$. Show that $B \subseteq \bigcap_{\alpha \in \Lambda} A_\alpha$.
2. Assume instead that $A_\alpha \subseteq B$ for all $\alpha \in \Lambda$. What can we conclude in this case? (Prove that your answer is correct.)

Problem 8. Complete Exercise 17 on page 198 of the textbook (*Mathematical Reasoning* by Sundstrom).

Problem 9. Do not turn in, but complete the progress checks in the sections of the textbook.